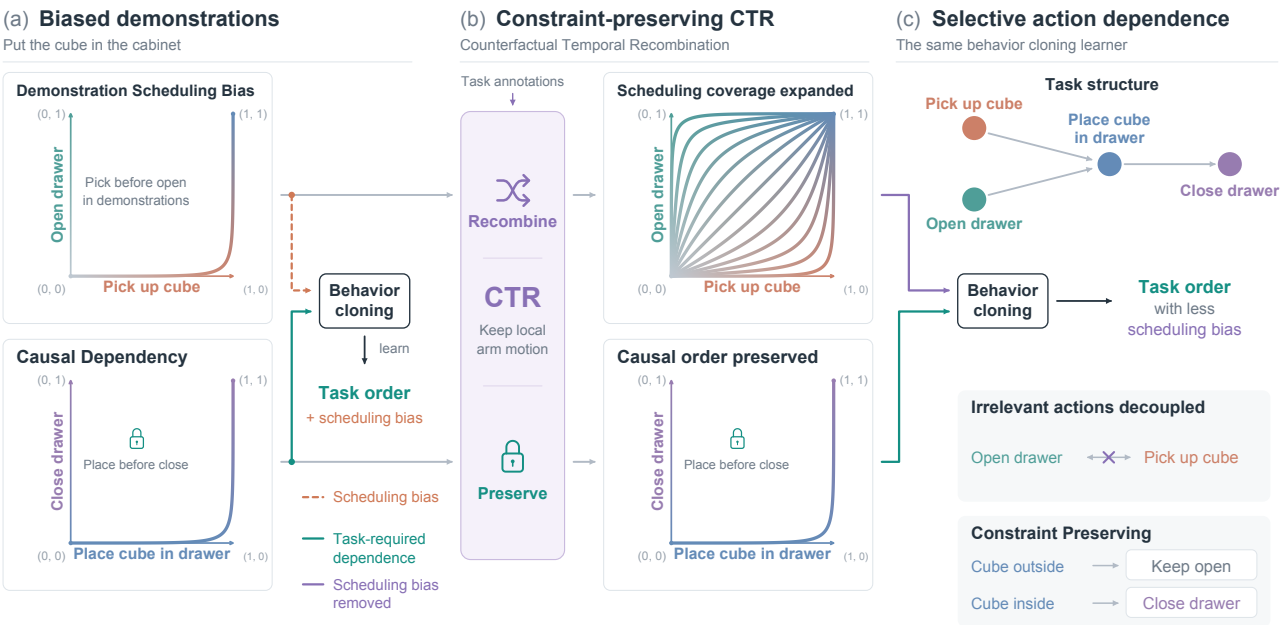

(a) Biased demonstrations
Put the cube in the cabinet
Demonstration Scheduling Bias
(0, 1)
(1, 1)
Open drawer
Pick before open in demonstrations
(0, 0)
Pick up cube
(1, 0)
Causal Dependency
(0, 1)
(1, 1)
Close drawer
Place before close
(0, 0)
Place cube in drawer
(1, 0)
Behavior cloning
learn
Task order
+ scheduling bias
Scheduling bias
Task-required dependence
Scheduling bias removed
(b) Constraint-preserving CTR
Counterfactual Temporal Recombination
Task annotations
Recombine
CTR
Keep local arm motion
Preserve
Scheduling coverage expanded
(0, 1)
(1, 1)
Open drawer
(0, 0)
Pick up cube
(1, 0)
Causal order preserved
(0, 1)
(1, 1)
Close drawer
Place before close
(0, 0)
Place cube in drawer
(1, 0)
(c) Selective action dependence
The same behavior cloning learner
Task structure
Pick up cube
Place cube in drawer
Close drawer
Open drawer
Behavior cloning
Task order
with less scheduling bias
Irrelevant actions decoupled
Open drawer
Pick up cube
Constraint Preserving
Cube outside
Keep open
Cube inside
Close drawer